

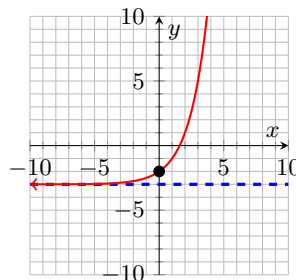
You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

1. Evaluate the following [4 pts each]:

- a. $-2^{-4} = -\frac{1}{2^4} = -\frac{1}{16}$
- b. $\log_{\pi} 1 = 0$
- c. $\ln(e^3) = 3$
- d. $\lim_{x \rightarrow -\infty} 7^x = 0$
- e. $\lim_{x \rightarrow \infty} \log_7 x = \infty$

2. Let $f(x) = 2^x - 3$.

- a. Graph $f(x)$.
- b. Find the domain and range of $f(x)$.

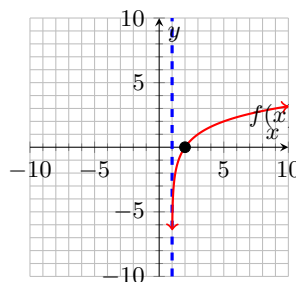


Domain: $(-\infty, \infty)$
 Range: $(-3, \infty)$

3. Let $g(x) = \log_4(x - 1)$.

- a. Graph $g(x)$.
- b. Find the domain and range of $g(x)$.

c.



Domain: $(1, \infty)$
 Range: $(-\infty, \infty)$

4. Use the properties of logarithms to expand the expression as much as possible.

- a. $\log_7(3y)$
 $= \log_7 3 + \log_7 y$
- b. $\log\left(\frac{10^6}{x}\right)$
 $= \log(10^6) - \log x = 6 - \log x$

5. Use the properties of logarithms to rewrite the expression as a single logarithm.

- a. $\log_c(16) - \log_c(2)$
 $\log_c 16/2 = \log_c 8$
- b. $\ln(2) + 6 \ln(x)$
 $\ln 2 + \ln x^6 = \ln(2x^6)$

6. Solve for x . Write your answer as an exact and approximate value rounded to three decimal places [4 pts each].

- a. $4^x = 3$

$$\begin{aligned} \ln 4^x &= \ln 3 \\ x \ln 4 &= \ln 3 \\ x &= \frac{\ln 3}{\ln 4} \approx .792 \end{aligned}$$

- b. $e^{2x+1} = 10$

$$\begin{aligned} \ln e^{2x+1} &= \ln 10 \\ 2x + 1 &= \ln 10 \\ x &= \frac{\ln 10 - 1}{2} \approx 1.803 \end{aligned}$$

c. $\log_6 x = 2$

$$\begin{aligned} 6^{\log_6 x} &= 6^2 \\ x &= 36 \end{aligned}$$

d. $\ln(40) = 3 - \ln(x)$

$$\begin{aligned} \ln 40 + \ln x &= 3 \\ \ln(40x) &= 3 \\ 40x &= e^3 \\ x &= \frac{e^3}{40} \approx .502 \end{aligned}$$

e. $\log_2(x+3) + \log_2(x-3) = 7$

$$\begin{aligned} \log_2(x^2 - 9) &= 7 \\ x^2 - 9 &= 7 \\ x^2 &= 16 \\ x &= \pm 4 \end{aligned}$$

Check, $x \neq -4$. So $x = 4$

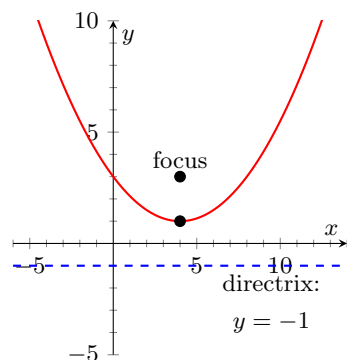
7. If \$500 is invested in an account at 5% annual interest compounded monthly, what is the amount in the account after 9 years?

$$500 \left(1 + \frac{.05}{12} \right)^{12 \cdot 9} \approx \$783.42$$

8. How long will it take \$500 to grow to \$1500 if it is invested at 10% compounded continuously?

$$\begin{aligned} 1500 &= 500e^{.10 \cdot t} \\ 3 &= e^{.10t} \\ \ln 3 &= .10t \ln e \\ \frac{\ln 3}{.10} &= t \approx 10.986 \text{ years} \end{aligned}$$

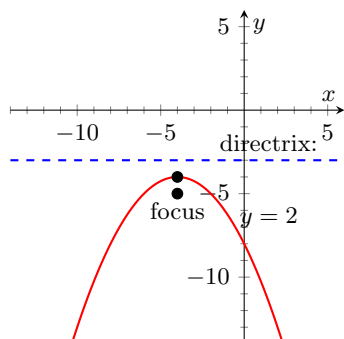
9. Find and graph the focus, directrix, and vertex of the parabola $y = \frac{1}{8}(x-4)^2 + 1$.



$$\frac{1}{8} = \frac{1}{4p} \Rightarrow p = 2.$$

vertex: $(4, 1)$
 focus: $(4, 1 + p) = (4, 3)$
 directrix: $y = 1 - 2 = -1$

10. Find and graph the equation of the parabola with focus $(-4, -5)$ and directrix $y = -3$. ($p = -1$)



$$p = \frac{(-5) - (-3)}{2} = -1.$$

So then the vertex is $(h, \text{directrix} + p) = (-4, -2)$
 $\Rightarrow y = -(x + 4)^2 - 2$